

ANNITIA Data Challenge: Solution Report

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Repository: github.com/astajd/annitia-final-submission

1. Executive summary

This report describes the final submission to the ANNITIA / Trustii.io / IHU ICAN data challenge. The task is to rank patients with MASLD by risk for two endpoints: major hepatic events and all-cause death. Submissions are scored by a weighted concordance index (C-index), $0.7 \cdot C_{\text{hepatic}} + 0.3 \cdot C_{\text{death}}$, which depends only on the ordering of the predicted risks. Because the metric is rank-based, the submitted predictions are emitted as pure ranks rather than calibrated probabilities or survival times.

The submitted solution is a deterministic ensemble that models the two endpoints separately and combines two independently developed modeling tracks. The hepatic endpoint uses a clinically motivated liver-stiffness gate between the two track anchors, with a disagreement-based fallback. The death endpoint uses a fixed-weight rank blend of two gradient-boosted survival models. Every combination parameter is fixed and the final assembly is fully deterministic.

The submitted file (referred to in the repository as `slot1_prediction.csv`) has 423 rows: one identifier column and two rank-valued risk columns. The repository provides two verification paths against this frozen artifact: a full from-raw retraining of every component, which reproduces the file byte-for-byte under a pinned environment, and a fast cached-output check that regenerates the submitted prediction from saved model-level outputs without raw data. Both have been run in the submitted repository.

This report presents only documented results and verification facts. It makes no claim about ranking, standing, or any final official score.

2. Challenge and evaluation

The challenge provides a longitudinal MASLD cohort with repeated clinical and non-invasive-test (NIT) measurements per patient, together with two time-to-event outcomes. The objective is to produce, for each test patient, a risk score for hepatic events and a risk score for death such that higher scores correspond to earlier/with-event patients under the C-index.

Why rank-valued outputs. The challenge evaluation uses the `scikit-survival` `concordance_index_censored` implementation. This metric depends on the ordering of predicted risks rather than their absolute calibration. The submitted pipeline therefore optimizes and emits rank-aggregated quantities throughout: blends are computed on `scipy.stats.rankdata` values

and the final columns are themselves ranks. Calibration is neither required nor attempted for this metric.

Data. The challenge dataset is synthetic. It contains static features and longitudinal `<variable>_v<i>` columns across repeated visits, plus the two outcomes in the training file. The test file has the same feature schema and no outcome columns. Because the data are synthetic, any feature-effect pattern observed during interpretation reflects the data-generating process rather than validated clinical biology; no causal claim is made anywhere in this work. The official files are present under `data/raw/` in this repository following organizer clarification (see `data/README.md`).

Submitted artifact. `frozen/slot1_prediction.csv` has 423 rows and the required schema `trustii_id`, `risk_hepatic_event`, `risk_death`, with both risk columns expressed as ranks. Its SHA-256 and MD5 checksums are recorded in `frozen/SHA256SUMS`; the MD5 is also reproduced in the verification table in Section 5.

A note on terminology: the repository refers to the two modeling tracks as **Track A** and **Track B**. These are internal labels for two independently developed modeling pipelines; the labels carry no meaning beyond that separation and are used consistently in the code and documentation.

3. Methodology

3.1 Two-track design

The solution combines two modeling tracks that were developed independently, each with its own preprocessing, feature engineering, cross-validation protocol, and survival models. Combining them at the end exploits the fact that independently constructed models tend to make partly uncorrelated errors, so a principled combination is more stable than either track alone. The two tracks supply *anchor* predictions for the hepatic endpoint; the death endpoint is built from a separate pair of survival models.

3.2 Track anchors (hepatic)

- **Track A anchor.** A 0.30/0.70 blend of a landmark random survival forest component and a permissive rank-mean component that itself combines additional random survival forest and XGBoost-Cox models. The 0.30/0.70 blend weight was selected by cross-validation (argmax of a mean-minus-standard-deviation criterion on a 5×10 CV grid), not tuned on the leaderboard; the exact component lineage is in `docs/PROVENANCE.md`.
- **Track B anchor.** A rank-blend of gradient-boosted survival models (LightGBM, CatBoost, XGBoost) trained across multiple prediction horizons, with blend weights pinned in a meta-data sidecar and selected from per-component out-of-fold (OOF) evaluation.

Both anchors are genuine survival-model ensembles rather than heuristic combinations.

3.3 Hepatic endpoint recipe

The hepatic risk is assembled from the two anchors in three deterministic steps:

1. **Liver-stiffness gate (12 kPa).** Each patient is routed by their most recent liver-stiffness (LS) measurement. Patients with $LS \geq 12$ kPa take their hepatic risk from the Track A anchor; patients with $LS < 12$ kPa, or with no available LS measurement, take it from the Track B anchor.
2. **Disagreement override (q95).** For each patient the normalized absolute rank difference between the two anchors is computed, $d = |\text{rank}(\text{Track B}) - \text{rank}(\text{Track A})| / N$. For the top 5% of patients by this disagreement (the q95 quantile), the gated value is replaced by a 50/50 rank-average merge of the two anchors. The rationale is that where two independently developed models disagree most strongly, a consensus is safer than committing to either single track.
3. **Rank transform.** The result is converted to ranks for the final hepatic-event risk column.

The 12 kPa value is a clinically motivated, interpretable liver-stiffness boundary used here as a MASLD/NIT high-risk or advanced-fibrosis threshold. It is **not** the Baveno VII rule-in threshold for clinically significant portal hypertension, which uses a higher liver-stiffness value (see Section 6 and reports/explainability/REFERENCES.md). The clinical-versus-statistical basis for choosing this threshold is detailed in Section 4.

3.4 Death endpoint recipe

The death risk is a fixed-weight rank blend of two gradient-boosted survival models, both trained on a 143-feature longitudinal-summary feature set:

- **CWGBSA:** scikit-survival ComponentwiseGradientBoostingSurvivalAnalysis, Cox loss, 300 estimators, learning rate 0.05, subsample 0.8, standardized features.
- **GBSA:** scikit-survival GradientBoostingSurvivalAnalysis, Cox loss, 200 estimators, learning rate 0.05, maximum depth 3, subsample 0.8, `max_features="sqrt"`.

Both were trained with `random_state = 42 + repeat` under a 5×10 stratified-by-event cross-validation protocol with mean-of-ranks aggregation. The final death risk is

$$\text{risk}_{\text{death}} = \text{rankdata}(0.85 \cdot \text{rank}(\text{CWGBSA}) + 0.15 \cdot \text{rank}(\text{GBSA})).$$

The 0.85/0.15 weighting was selected by OOF grid search over candidate ratios, not by leaderboard comparison; the componentwise model dominated in OOF evaluation, which is reflected in its larger weight.

3.5 Final prediction architecture

The figure below summarizes the two endpoint pipelines as they run at execution time. Every step shown is a fixed, deterministic operation; the selection history behind these choices is discussed separately in Section 4.

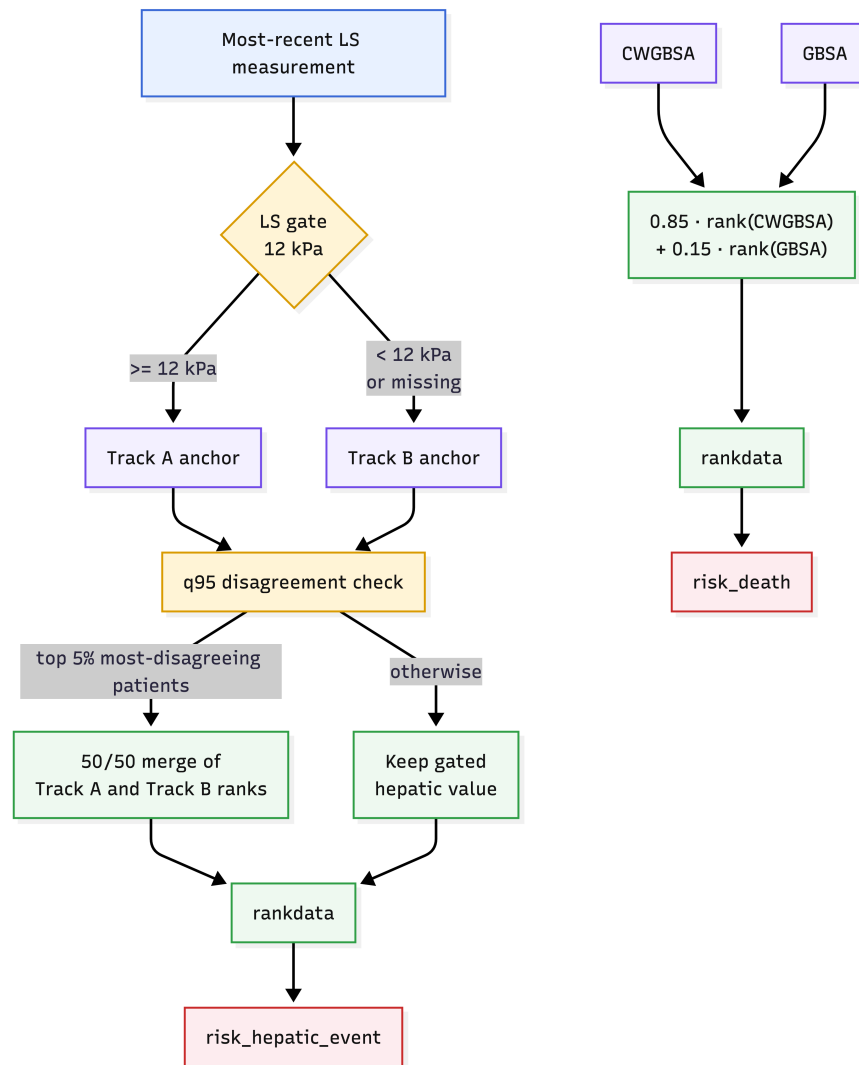


Figure 1: Final prediction architecture. The hepatic endpoint uses a liver-stiffness gate followed by a q95 disagreement override; the death endpoint uses a fixed 0.85/0.15 rank blend.

3.6 Final assembly

Both endpoint risks are converted to ranks for submission, consistent with the rank-based objective. The output is ordered by `trustii_id` ascending. The final assembly logic (`build_slot1_only.py`) is the same code path used both for the fast verification (Path B) and, over freshly regenerated components, for the full from-raw retraining (Path A).

4. Selection discipline and leakage controls

A central aim of this submission is to be exact about *how* each decision was made (by cross-validation, by clinical reasoning, or by reference to the public leaderboard) rather than to present a uniformly clean narrative. The following table summarizes the final-recipe decisions; the prose below it covers the one decision that requires care.

Decision	Selection method	Notes
q95 disagreement-override threshold	Cross-validation (OOF)	Grid search over quantiles {0.75, 0.85, 0.90, 0.92, 0.95, 0.97, 0.99}; not leaderboard-tuned.
0.85/0.15 death-model blend	Cross-validation (OOF)	Grid search over ratios; componentwise model dominated in OOF.
Track A anchor blend weight (0.30/0.70)	Cross-validation (OOF)	Argmax of mean-std on a 5×10 CV grid; leaderboard values appear only as post-selection reference metadata, not as inputs.
Track B anchor blend weights	Cross-validation (OOF)	Per-component OOF evaluation; weights pinned in a metadata sidecar.
Liver-stiffness gate threshold (12 kPa)	Clinical reasoning	The one final-recipe parameter not chosen by OOF optimization; see below.
Primary submission among LS=12 candidates	Public leaderboard	Choice among candidates that differ in death blend / override, not the gate threshold; see below.

4.1 The gate threshold and the primary-candidate choice are two separable decisions

These two decisions are easy to conflate, so they are stated separately and precisely.

The 12 kPa gate threshold was a clinical choice. Out-of-fold evaluation marginally favored an 18 kPa gate, by about +0.0012 weighted-C. The OOF-best candidate, referred to in the repository as `finalprobe_1`, is identical to the submitted candidate, `finalprobe_3`, in every component except the gate threshold. Both use the q95 disagreement override and the same 0.85/0.15 death

blend; the only difference is the gate threshold, 18 kPa instead of 12 kPa. The 12 kPa threshold was retained for clinical defensibility, knowingly accepting that small out-of-fold cost. The 18 kPa variant was OOF-confirmed only and was never scored on the public leaderboard.

The primary submission was selected within the LS=12 family using the public leaderboard. Among the submitted candidates (all sharing the 12 kPa gate and differing only in the death blend and the disagreement override, not in the gate threshold), the public leaderboard informed which configuration was submitted as primary. The submitted candidate `finalprobe_3` had a documented public-LB score of 0.92198 on the now-closed board. This leaderboard step did not concern the gate threshold.

In short: most parameters were OOF-selected, the gate threshold was a deliberate clinical choice (over an OOF-best 18 kPa gate that was never leaderboard-scored), and the public leaderboard informed only the primary-submission choice within the LS=12 candidate family. The 0.92198 value is reported here strictly as that candidate's public-leaderboard score in the selection/provenance context; it is not a final official score, rank, or standing.

4.2 Feature and target leakage controls

- **No private or hidden labels.** No private leaderboard labels or held-out outcome information were used at any point.
- **No manual patient-level intervention.** No predictions were hand-edited, reordered, or patched against any reference file. The submitted file is the deterministic output of the assembly recipe.
- **Leakage-prone feature constructions were audited and excluded.** During development, time-aligned and censoring-aware feature variants were tested. Variants that encoded target timing or post-outcome information produced implausibly strong internal validation but did not generalize, and were excluded from the final selected solution.
- **Cached intermediates are predictions, not data.** The cached files consumed by the fast verification path are saved model-level prediction outputs, not raw challenge data, and are labeled as such.

One model-behavior finding on the death endpoint warrants disclosure as a generalization caveat rather than as concealed leakage; it is described in Section 6 and revisited in Section 7. Full detail is in `docs/LEAKAGE_AUDIT.md`.

5. Reproducibility

The frozen submission file is the authoritative artifact, and the repository provides two independent ways to verify it. Both have been run in the submitted repository.

5.1 Path A: full raw-data retraining

```
bash pipeline_full/runnable/retrain_all_from_raw.sh
```

This single command retrains **every** model component from the raw challenge data in `data/raw/` (the Track A and Track B hepatic anchors, the 12 kPa gate, the 50/50 merge, and the CWGBSA / GBSA death models), and then runs the deterministic final assembly over the freshly regenerated components. It does not read the cached intermediates as model inputs.

In the submitted repository, the full retraining output is **byte-for-byte identical** to the frozen submitted file. The verification result is summarized below.

Check	Result
Submitted file	<code>frozen/slot1_prediction.csv</code>
Rows	423
MD5 match	<code>fb04658b99f89ec822e9c604d537dcae</code>
Endpoint agreement	rank-identical and float-exact for both endpoints
Runtime	approximately 41 minutes on the tested host

Byte-exactness is environment-pinned. It is verified under the pinned environment recorded in the retraining requirements file (Python 3.11.4; numpy 2.4.4, pandas 2.3.3, scipy 1.17.1, scikit-learn 1.6.1, scikit-survival 0.27.0, xgboost 3.2.0, lightgbm 4.6.0, catboost 1.2.10, PyYAML 6.0.3) with the orchestrator's thread caps applied. The upstream survival models are stochastic and library-version sensitive, so byte-exact reproduction is claimed under this pinned environment; reruns under materially different library versions may produce close but not byte-identical outputs. The thread caps are functional, not cosmetic: without them the survival forests over-subscribe threads and thrash. Stage-by-stage detail is in the retraining documentation (`docs/RETRAINING.md`).

5.2 Path B: fast cached-output verification

Path B is the fast cached-output verification. It runs `frozen/verify.sh`, rebuilds the submitted prediction with `build_slot1_only.py`, and checks it with `validate_slot1_only.py`. It does not retrain models and does not read raw data. The rebuild applies the full assembly recipe (the 12 kPa gate read from cache, the q95 disagreement override with 50/50 fallback, and the 0.85/0.15 death blend) to the saved model-level outputs, and the verifier confirms that the regenerated prediction is rank-identical to the frozen submitted file. This path consumes only the cached prediction outputs and is independent of whether `data/raw/` is present.

5.3 Relationship between the two paths and the closed leaderboard

Path A demonstrates that the submitted file can be regenerated from raw data; Path B is a fast integrity check on the deterministic assembly recipe. Because the public leaderboard is now closed, a regenerated alternative cannot be re-scored on it, which is precisely why the frozen submitted file is treated as the authoritative artifact and all reproduction claims are stated relative to it.

6. Explainability

6.1 Structural interpretation is the most faithful explanation

The clearest explanation of the submission is structural, because the final recipe is a small, fully specified set of deterministic operations on component model outputs. For any patient a reviewer can read the exact decision path: which anchor supplied the hepatic value (via the 12 kPa gate), whether that value was overridden (top-5% disagreement $\rightarrow$ 50/50 merge), and how the death rank was composed (0.85 CWGBSA + 0.15 GBSA). This structural account does not rely on attribution heuristics and is given in full in Sections 3 and 4.

A consequence of this structure is that per-component feature attributions do **not** explain the final hepatic output. The gated hepatic ensemble is a discrete switch followed by a top-5% override to a consensus merge, not an additive combination, so standard additive attribution methods have no faithful decomposition of it. Attributions computed on the individual anchors describe those anchors in isolation, not the gated rank. This is stated rather than worked around.

6.2 The 12 kPa gate: clinical grounding and the Baveno disclaimer

The 12 kPa gate is a clinically interpretable threshold, with its wording matched to the literature. It is used here as a MASLD/NIT high-risk or advanced-fibrosis liver-stiffness threshold: under the AASLD 2023 NAFLD practice guidance, vibration-controlled transient elastography liver stiffness below 8 kPa rules out advanced fibrosis, 8–12 kPa is indeterminate, and above 12 kPa is associated with a high likelihood of advanced fibrosis. The 12 kPa gate aligns with the upper boundary of that three-tier scheme.

This is explicitly **not** the Baveno VII threshold for clinically significant portal hypertension. Baveno VII uses a “rule of five” (10, 15, 20, 25 kPa), with ≥ 25 kPa as the rule-in for clinically significant portal hypertension; it is cited only for those values, not as support for the 12 kPa gate. The citation-to-claim mapping is in `reports/explainability/REFERENCES.md`. As noted in Section 4, out-of-fold evaluation marginally favored a higher threshold and 12 kPa was retained for clinical defensibility.

6.3 The death endpoint: a follow-up-window signal, disclosed

The death endpoint is where post-hoc attribution is most feasible, and it produced the most important finding in this analysis. The two death components were interpreted by their model-native importances (sparse coefficients for CWGBSA, impurity importances for GBSA) from a single seed-42 reference fit; no retraining of the cross-validation ensemble was done and the cached predictions are unchanged.

The componentwise model (CWGBSA, weight 0.85) is intrinsically sparse: only two features receive nonzero coefficients, `Age_slope` (coefficient magnitude 0.53) and `Age_rel_delta` (0.38). Both are age-trajectory and follow-up-window features rather than disease-physiology markers: `Age_slope` effectively encodes whether a patient has two or more visits, and `Age_rel_delta` encodes the observed visit-window length. This must be read as model behavior on a synthetic survival dataset, not as a clinical mortality mechanism: on this cohort, the death endpoint is largely predictable from follow-up-duration-correlated features rather than from biomarker physiology.

An internal audit (reports/explainability/audit_CWGBSA_death_model.md) corroborates this: an age-features-only model matches the full model's out-of-fold C-index (**0.9704 versus 0.9703**), and a single follow-up-duration feature reaches **0.969**. This is not concealed target leakage in the sense of directly using event or censoring columns as features; it is a synthetic-cohort follow-up-window signal that is informative here and that transfers to the test split. Its practical consequence, that this discrimination may not transfer to a cohort with a different follow-up structure, is recorded as a limitation in Section 7.

The gradient-boosting model (GBSA, weight 0.15) is denser. Its top impurity importances are still dominated by age-trajectory summaries, with biomarker summaries such as FibroTest and liver-stiffness measures appearing lower. The same follow-up-window caveat applies, with somewhat more biomarker contribution than CWGBSA. Impurity importances are biased toward high-cardinality continuous features and should be read as a relative ranking, not as effect sizes.

All explainability material is post-hoc: it interprets the submitted models after the fact and did not influence model construction, model selection, or the submitted predictions. The supporting tables, scripts, and references are under reports/explainability/.

7. Limitations

- **Synthetic data; no causal clinical claims.** The dataset is synthetic, so feature-effect patterns reflect the data-generating process rather than validated clinical associations. None of the interpretation in Section 6 should be read as causal or clinically established.
- **Byte-exact reproduction is environment-pinned.** Path A reproduces the submitted file byte-for-byte under the pinned environment in requirements_retrain.txt with the thread caps applied; the survival models are stochastic and library-version sensitive, so reruns under materially different versions may differ.
- **Primary-slot selection was public-leaderboard-informed (within the LS=12 family).** The submitted candidate was chosen with reference to the public leaderboard among candidates that share the 12 kPa gate and differ only in the death blend and the disagreement override. The within-recipe parameters (q95 override, death blend, anchor weights) were cross-validation-selected, and the gate threshold was a clinical choice over an OOF-best 18 kPa variant that was never leaderboard-scored.
- **Architecture-level choices were not exhaustively ablated.** The final hepatic recipe combines a liver-stiffness gate, a disagreement override, and rank merging. Individual parameters were documented and selected by cross-validation, clinical reasoning, or public-leaderboard comparison as described above, but the broader architecture was not presented as the result of an exhaustive search over all possible ensemble structures.
- **The death endpoint's discrimination rests on a follow-up-window signal.** As shown in Section 6, the dominant death component is driven by follow-up-window / age-trajectory features (Age_slope, Age_rel_delta); an age-features-only model matches the full model's OOF C-index (0.9704 vs 0.9703) and a single follow-up-duration feature reaches 0.969. On this synthetic cohort the signal is real and transfers to the test split, but the endpoint's strong discrimination may not transfer to clinical data with a different censoring / follow-up structure.
- **The closed leaderboard.** No regenerated alternative can be re-scored on the now-closed public leaderboard, which is why the frozen file is the authoritative artifact.
- **Revalidation before any clinical use.** Faithful attribution for the gated hepatic prediction would require methods that account for the gating and override logic rather than per-model

attribution, and any clinical claim would require validation on real rather than synthetic data. Both are out of scope for this challenge submission.

8. Conclusion

The final submission is a deterministic two-track ensemble: a clinically gated, disagreement-aware combination of two independently developed survival-model anchors for the hepatic endpoint, and a fixed-weight rank blend of two gradient-boosted survival models for the death endpoint. Selection discipline is documented decision by decision: most parameters cross-validation-selected, the 12 kPa gate a deliberate clinical choice, and the public leaderboard used only to pick the primary submission within the LS=12 candidate family. The submitted file is verifiable two ways: a full from-raw retraining that reproduces it byte-for-byte under a pinned environment, and a fast cached-output check that confirms the deterministic recipe without raw data. The principal interpretive finding, that the death endpoint's discrimination rests on a synthetic-cohort follow-up-window signal, is disclosed openly, along with its generalization limit. The intent throughout is an honest, reproducible, and transparently documented submission.

Full methodological, provenance, reproducibility, leakage-audit, explainability, and limitation details are provided in the repository documentation under docs/, with citation-to-claim mapping under reports/explainability/REFERENCES.md.